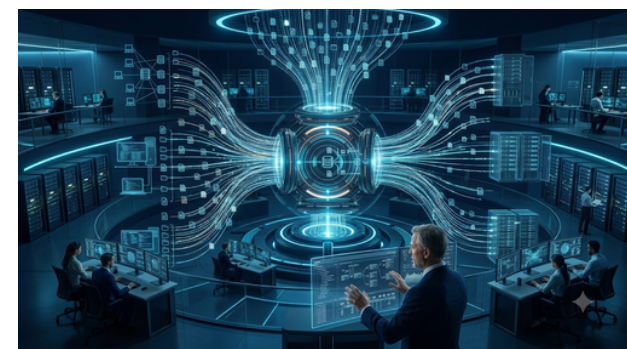


SMART INDIA HACKATHON 2026

- Problem Statement ID : **SIH26148**
- Problem Statement Title : **JOCKY - Next-Gen Forensic Analysis Language & Framework**
- Theme : **Blockchain & Cybersecurity**
- PS Category : **Software**
- Team ID : **2E9A7E**
- Team Name : **Bhole Chature**

PROBLEM EXISTING

- **Tool Fragmentation:** Forensic investigations depend on multiple specialized tools.
- **Low-Level Complexity:** Investigators require platform-specific knowledge and APIs.
- **Multi-System Challenge:** Investigations across different authorized systems are difficult to orchestrate.



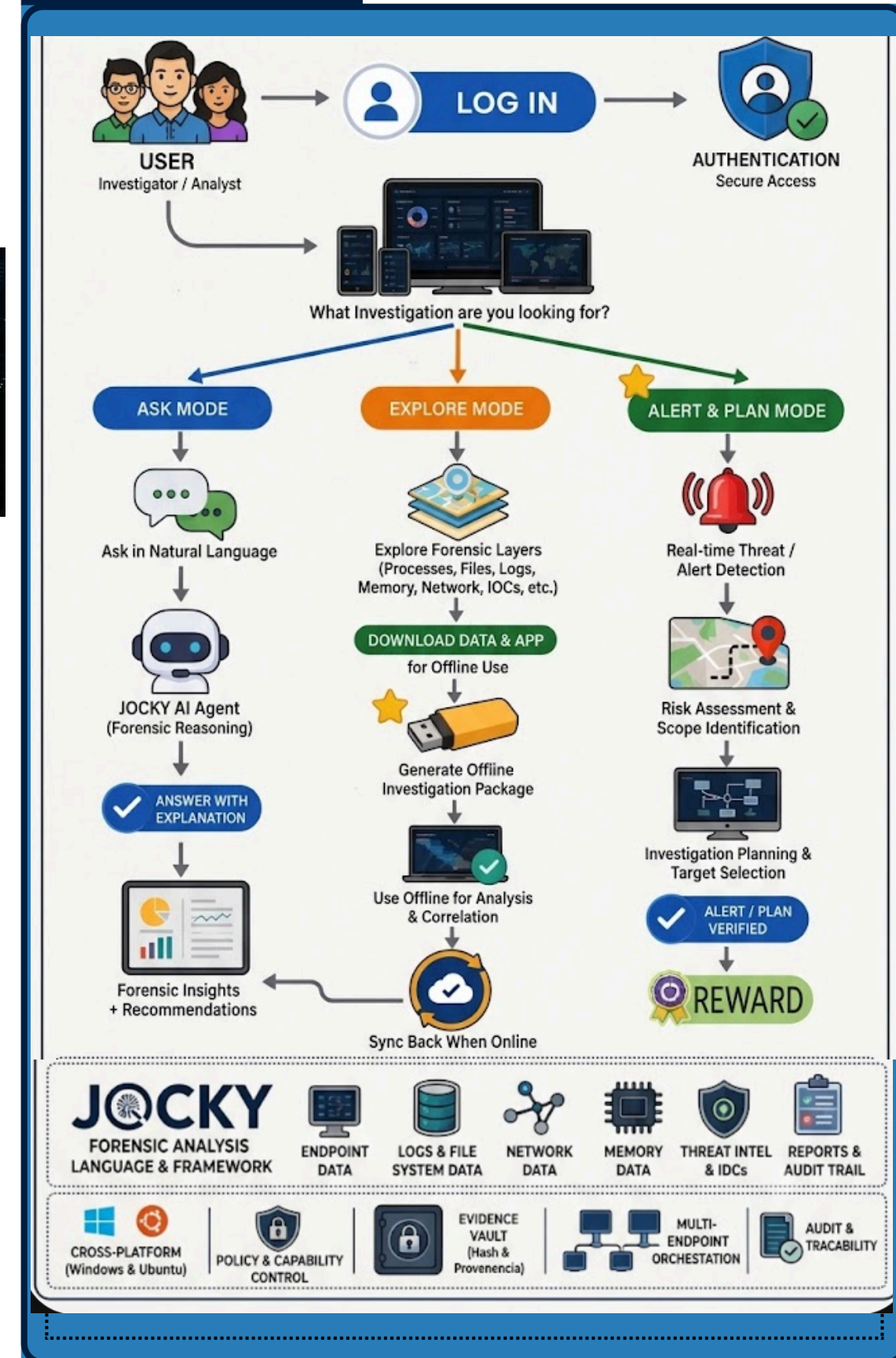
PROPOSED SOLUTION

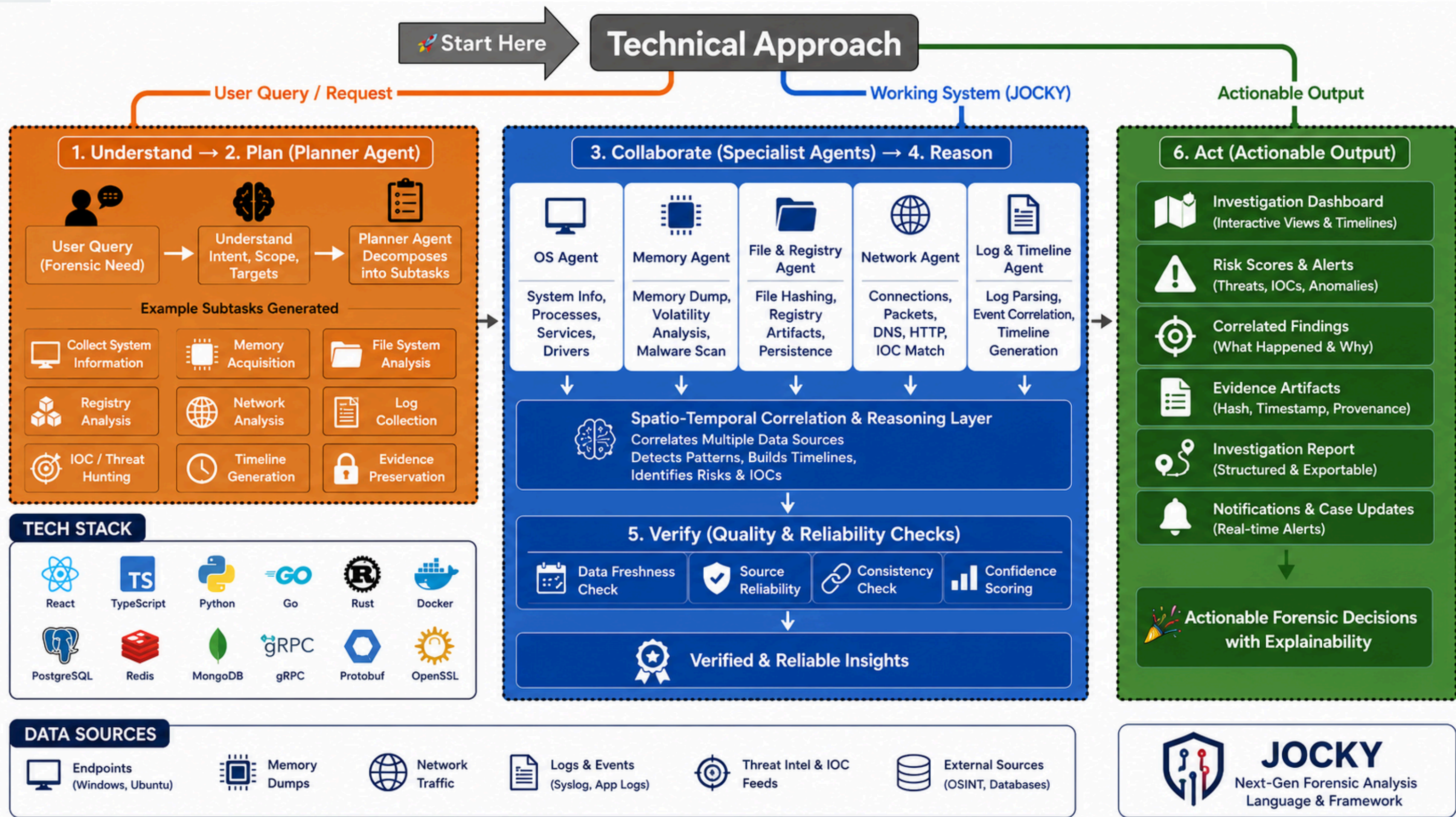
- **Forensic DSL:** Write reusable forensic investigation logic using **simple forensic primitives**.
- **Cross-Platform Runtime:** Execute the same investigation workflow across **Windows and Ubuntu**.
- **Controlled Execution:** Enforce authorization, capabilities, scope and execution policies.
- **Evidence Vault:** Record hashes, timestamps, provenance and execution metadata.

UVP (UNIQUE VALUE PROPOSITION)

- **Write Once, Investigate Everywhere:** One language for multiple authorized platforms.
- **Unified Forensic Intelligence:** Correlate endpoint, memory, filesystem and network artifacts.
- **Evidence-Aware Execution:** Produce traceable and verifiable forensic results.
- **Multi-Endpoint Investigation:** Orchestrate investigations across multiple authorized systems.

ARCHITECTURE





FEASIBILITY :**Technical :**

Compiler Architecture – Lexer, Parser, AST, IR & Runtime provide a practical foundation for JOCKY.

Innovation : Forensic DSL + Platform-Neutral IR enables reusable and programmable forensic investigations.

Operational : Policy-controlled execution, distributed agents & centralized orchestration support authorized multi-endpoint investigations.

Economic : Modular & Scalable – Reusable workflows and open-source components can reduce repetitive tooling and development effort.

TECHNICAL CHALLENGES :

Risk : Operating-system differences can make the same forensic workflow difficult to execute consistently.

Mitigation : Use platform-specific runtime adapters behind a common platform-neutral IR.

VIABILITY :

Adoption : Growing Cyber-Forensics Need – Increasing demand for scalable, repeatable and verifiable digital investigations.

Government & Enterprise : Potential use in authorized government, defence, SOC, CERT & enterprise forensic environments.

Security & Trust : Evidence Integrity – Hashes, timestamps, provenance and execution metadata improve forensic traceability.

Scalability & Retention : Multi-Endpoint Platform – Architecture can expand from a single system to large-scale authorized investigations.

BUSINESS / USER CHALLENGES :

Risk : Investigators may find complex forensic outputs difficult to interpret and correlate.

Mitigation : Provide centralized dashboards, correlated findings, evidence references and clear forensic reports.

- **The Economic & Operational Impact:** Reduces repetitive forensic work, improves investigation efficiency and enables faster incident response.
→ Helps investigators analyze multiple systems using reusable workflows.
- **Security & Forensic Gains:** Provides controlled, repeatable and verifiable forensic investigations with policy-based execution.
→ Improves evidence integrity through hashes, timestamps and provenance.
- **Scalability & Governance Transformation:** Enables centralized investigation across multiple authorized endpoints with auditable execution.
→ Supports stronger enterprise, SOC and government cyber-forensic capabilities.

JOCKY vs Existing Solutions		
Feature	Existing Solutions	JOCKY
 Forensic DSL (Domain Specific Language)	✗	✓
 Cross-Platform (Windows & Ubuntu)	✗	✓
 Platform-Neutral Intermediate Representation (IR)	✗	✓
 Policy-Based & Controlled Execution	⚠	✓
 Multi-Endpoint Orchestration	⚠	✓
 Evidence Integrity & Provenance	⚠	✓
 Correlated Forensic Insights & Reporting	⚠	✓
 Auditability & Tamper Evident Logs	✗	✓

HOW JOCKY WORKS

MODE 1: ASK — Forensic Intelligence

- **For:** Investigators needing forensic information or decision support.
- **Process:** Ask query → JOCKY analyzes → Correlate evidence → Explain result.
- **Time:** Seconds–minutes.
- **Best For:** Process, filesystem, logs, memory & network investigations.

MODE 2: EXPLORE — Forensic Analysis

- **For:** Investigators exploring system and forensic artifacts.
- **Process :** Select target → Choose artifacts → Analyze → Correlate → Verify.
- **Layers :** Processes, files, logs.
- **Best For :** Offline and post-incident investigations.

MODE 3: ALERT — Proactive Forensics

- **For :** Detecting emerging security events across authorized endpoints.
- **Process :** Monitor data → Detect indicators → Assess findings → Alert investigator.
- **Output :** Finding + affected host + evidence + recommended action.
- **Best For :** Multi-endpoint investigations, SOCs

Research & References

- Digital Forensics & DFIR – Research on forensic investigation, incident response and digital evidence analysis.
- Compiler Design – Lexer, parser, AST and Intermediate Representation for building the JOCKY forensic DSL.
- Cross-Platform Forensics – Windows and Ubuntu runtime approaches for consistent forensic investigation workflows.
- Evidence Integrity & Provenance – Cryptographic hashing, timestamps and audit trails for verifiable forensic results.
- Endpoint & Network Forensics – Processes, filesystem, logs, memory, network connections and IOC analysis.
- Forensic Automation – Reusable investigation workflows and centralized multi-endpoint orchestration.

SOLUTIONS ALREADY EXIST

- Existing forensic platforms provide specialized capabilities for filesystem, memory, endpoint and network analysis.
- Available Solutions: Autopsy / The Sleuth Kit / Volatility / KAPE / Velociraptor / osquery / YARA / Sigma

OUR SOLUTIONS STANDS OUT

- Programmable: Forensic DSL
- Cross-Platform: Windows + Ubuntu
- Controlled: Policy-based execution
- Verifiable: Hashes + provenance
- Scalable: Multi-endpoint orchestration

PROJECT RESOURCES

